

MATH 340

Quiz 5

March 14, 2025

(20 points) Find the eigenvalues and associated eigenvectors of the following matrices, show your work.
Only work inside the box will be graded.

1. $\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ Hint: they are real.

2. $\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$ Hint: they are complex.

3. $\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 3 \end{bmatrix}$ Hint: they are real.

4. $\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ Hint: they are complex.